MACHINE LEARNING · FINAL PROJECT

Ensemble Methods

Under what conditions does boosting outperform bagging, and vice versa — and why?

Boosting vs. *Bagging*

From-scratch implementation of Decision Trees, AdaBoost, and Random Forest,
with an empirical comparison supported by PCA / K-Means / DBSCAN analysis

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AI Academy · National AI Center
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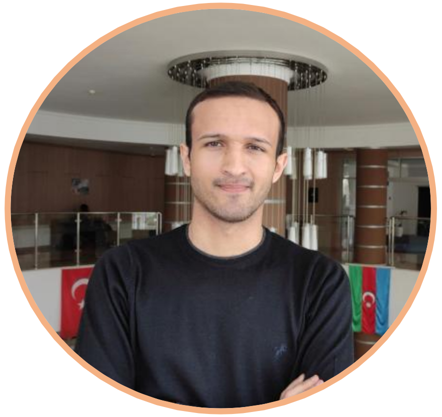
Team Contributions

Ferman Khankishiyev

Team Lead

CI
Decision Tree
Testing, code quality
Multiclass part for AdaBoost
Metrics(f1, auc-roc, accuracy)
Head-to-Head Comparison experiment
Baseline experiment



Bayram Bayramov

Developer Lead

AdaBoost
Preprocessing
DT&RF-optimization
Exploratory Data Analysis
Adaboost scaling experiment
Noise Robustness experiment

Mahmud Ramazanov

Developer Support

Presentation
Random Forest
Random Forest scaling experiment
Bias-variance decomposition experiment

Murad Taghizade

Documentation Lead
Developer Support

Presentation
Report
PCA/DBSCAN/K-Means
Bias-variance Decomposition
Unsupervised Analysis

METHODOLOGY

Three Models Implemented From Scratch

1 · Decision Tree

CART methodology, Gini / Entropy criterion, depth-based pruning, exhaustive search over continuous features.

2 · AdaBoost (SAMME)

Depth-1 decision stumps as weak learners, sample-weight update rule, staged predictions.

3 · Random Forest

Bootstrap aggregating, random feature subsampling, out-of-bag (OOB) accuracy, parallel training.

sklearn's DecisionTreeClassifier / AdaBoostClassifier / RandomForestClassifier are used only as a sanity-check baseline.

Decision Tree — CART Algorithm

Splitting Criterion

Gini impurity

$$I_G(\mathrm{p}) = 1 - \sum_{\mathrm{c}=1}^{\mathrm{C}} p_c^2$$

Entropy

$$I_E(\mathrm{p}) = -\sum_{\mathrm{c}=1}^{\mathrm{C}} p_c \cdot \log_2(p_c)$$

Impurity reduction

$$\Delta \mathrm{I} = \mathrm{I}_{\mathrm{parent}} - N_{left} / \mathrm{N} \cdot \mathrm{I}_{left} - \mathrm{N}_{right} / \mathrm{N} \cdot \mathrm{I}_{right}$$

Stopping conditions: max_depth reached, min_samples_split violated, a pure node, or identical feature vectors.

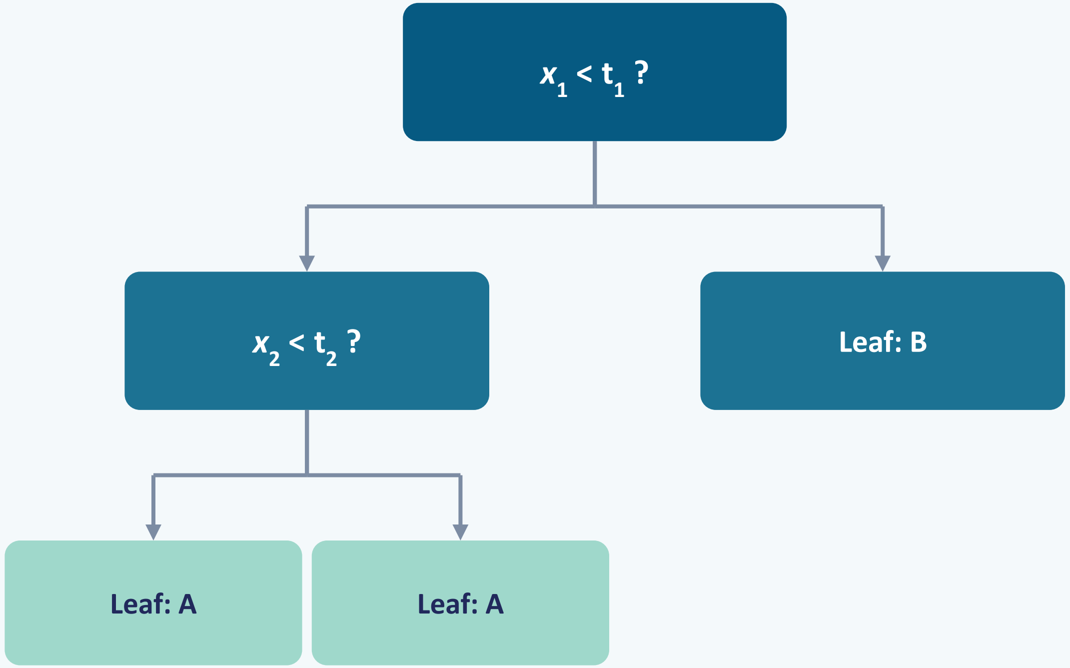


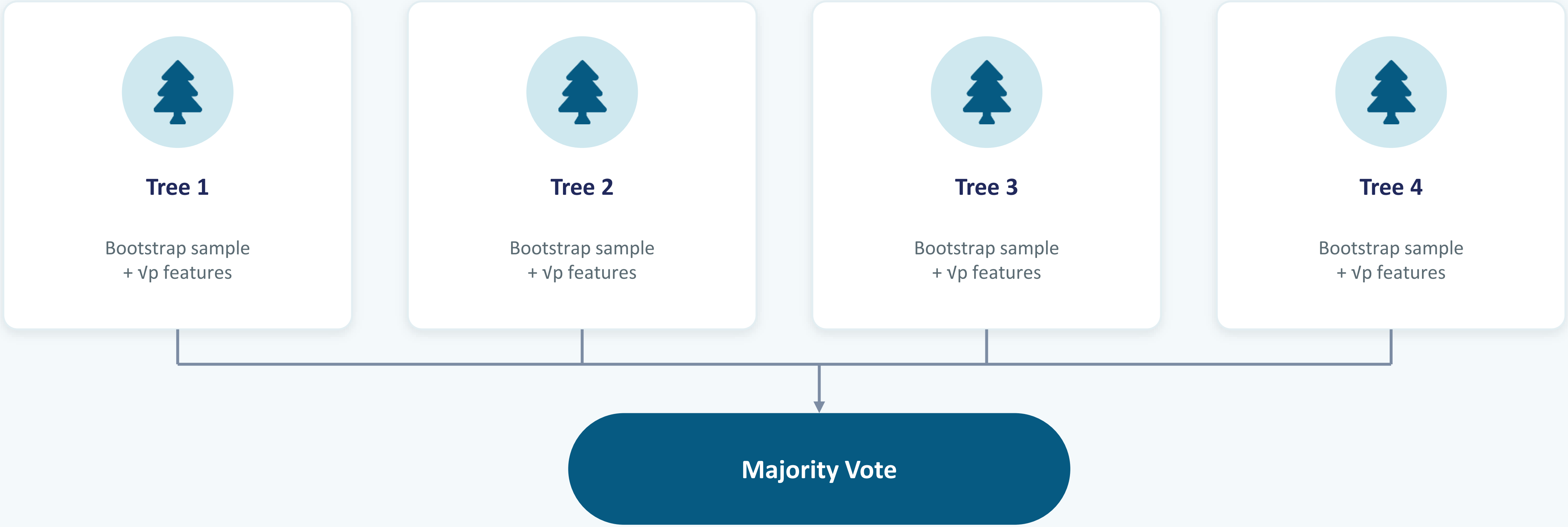
Figure: a two-level decision-tree example

AdaBoost — Weight Update Rule

1. Initialize equal weights: $w_i = 1 / \mathrm{N}$
2. Fit a weak learner (stump) h_m on the weighted data
3. Compute weighted error: $\varepsilon_m = \Sigma\ w_i \cdot 1[h_m(x_i) \neq y_i] / \Sigma\ w_i$
4. Model weight: $\alpha_m = \ln((1 - \varepsilon_m) / \varepsilon_m)$
5. Update sample weights: $w_i \leftarrow w_i \cdot \exp(\alpha_m \cdot 1[\mathrm{error}])$, then normalize
6. Final prediction: $\hat{y} = \mathrm{argmax}_{\mathrm{k}}\ \Sigma\ \alpha_m \cdot 1[h_m(\mathrm{x}) = \mathrm{k}]$

If $\varepsilon_m \geq 0.5$ the algorithm stops; if $\varepsilon_m = 0$ it is clipped to 10^{-10}.

Random Forest — Bagging + Feature Subsampling



OOB (Out-of-Bag) evaluation: for each sample, predictions are pooled only from the trees that never saw it — model generalization is measured without a separate test set.

Feature importances: averaged impurity reduction across all trees.

Unsupervised Analysis Pipeline

PCA

Scree plot, cumulative explained variance, projection onto the first 2 components.

K-Means

Elbow method (k = 1–10), 10 restarts, Adjusted Rand Index (ARI).

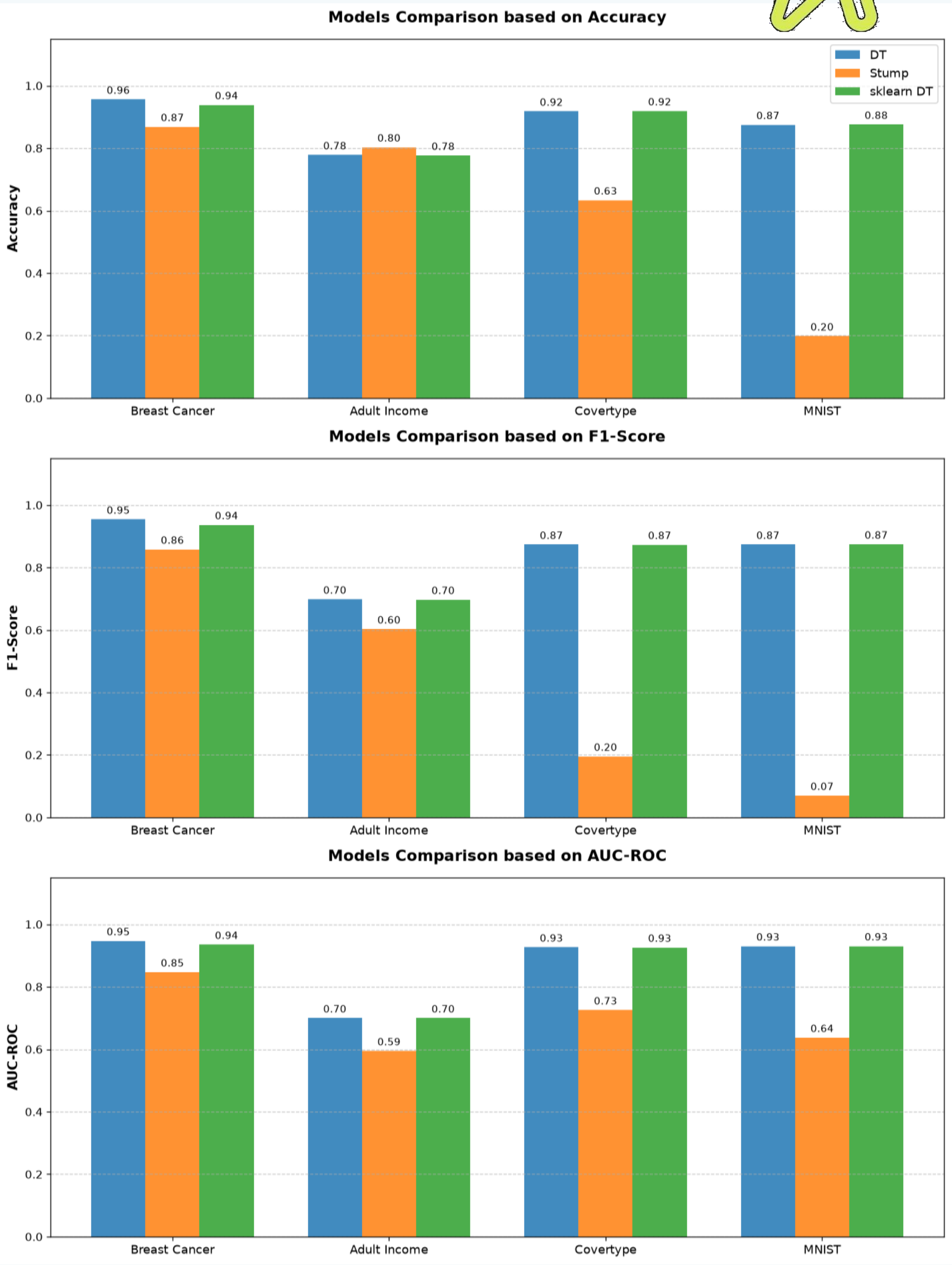
DBSCAN

k-distance plot, core/noise point classification, ARI and noise fraction.

2D PCA scatter plots colored by true class, K-Means clusters, and DBSCAN clusters are compared side by side.

Baseline

- Train a single unpruned DecisionTree on an 80/20 train/test split.
- Train a depth-1 stump DecisionStump.
- Train sklearn.tree.DecisionTreeClassifier with identical parameters.
- Report accuracy, F1-macro, and AUC-ROC for binary or macro-averaged AUC for multi-class.



1) Breast Cancer

Model	Accuracy	F1-score	AUC
DecisionTree (ours)	0.9561	0.9539	0.9468
DecisionStump	0.8684	0.8579	0.8468
sklearn Tree	0.9386	0.9364	0.9351

Accuracy difference vs sklearn: 1.87% (PASS)

2) Adult Income

Model	Accuracy	F1-score	AUC
DecisionTree (ours)	0.7783	0.6981	0.6997
DecisionStump	0.8033	0.6027	0.5945
sklearn Tree	0.7763	0.6969	0.6996

Accuracy difference vs sklearn: 0.25% (PASS)

3) Covertype

Model	Accuracy	F1-score	AUC
DecisionTree (ours)	0.9178	0.8733	0.9265
DecisionStump	0.6335	0.1951	0.7254
sklearn Tree	0.9189	0.8729	0.9258

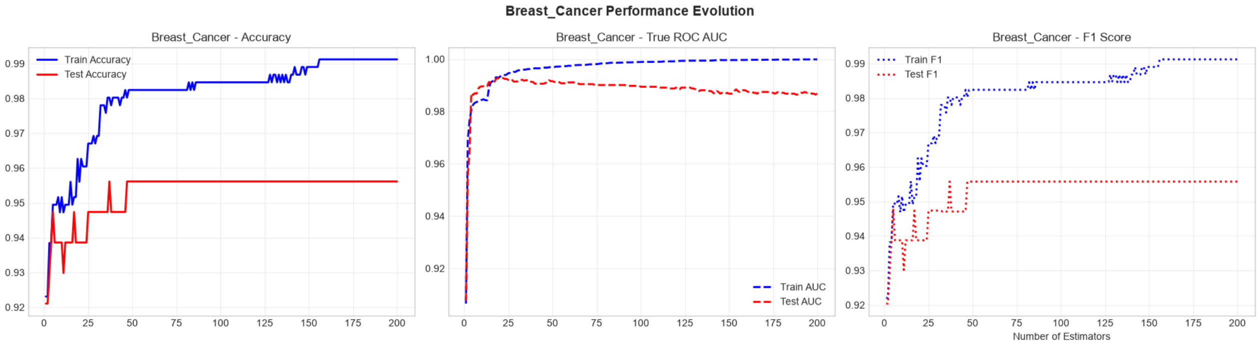
Accuracy difference vs sklearn: 0.13% (PASS)

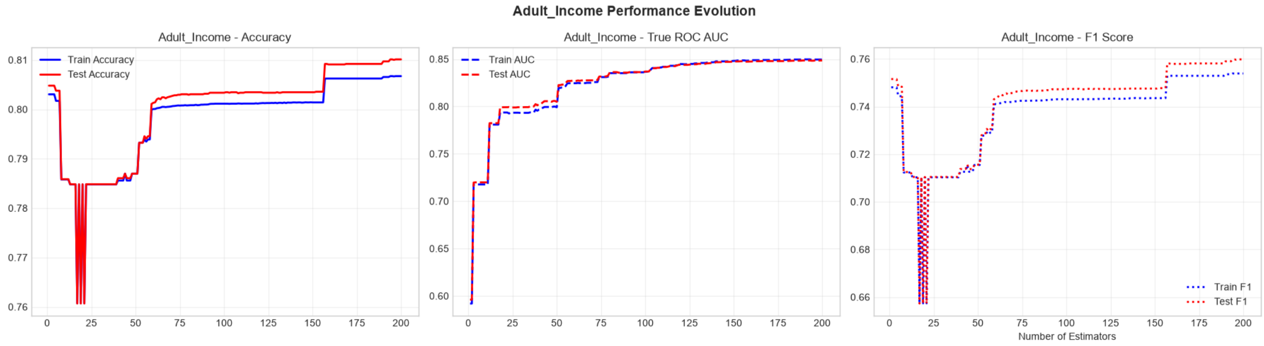
4) MNIST

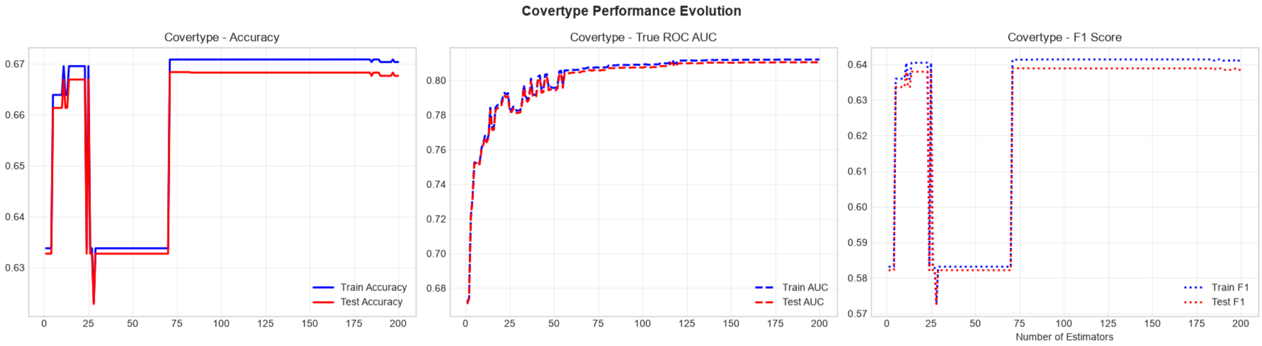
Model	Accuracy	F1-score	AUC
DecisionTree (ours)	0.8749	0.8734	0.9298
DecisionStump	0.1982	0.0697	0.6371
sklearn Tree	0.8758	0.8742	0.9302

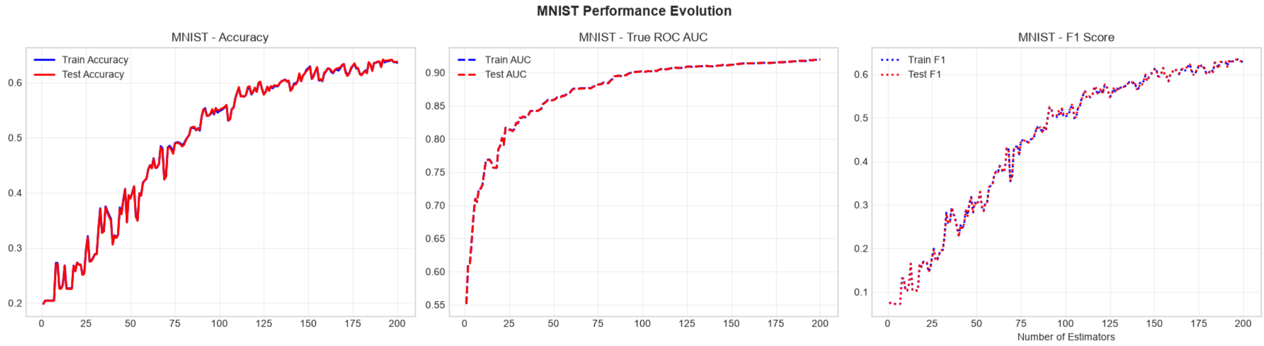
Accuracy difference vs sklearn: 0.10% (PASS)

Adaboost Scaling

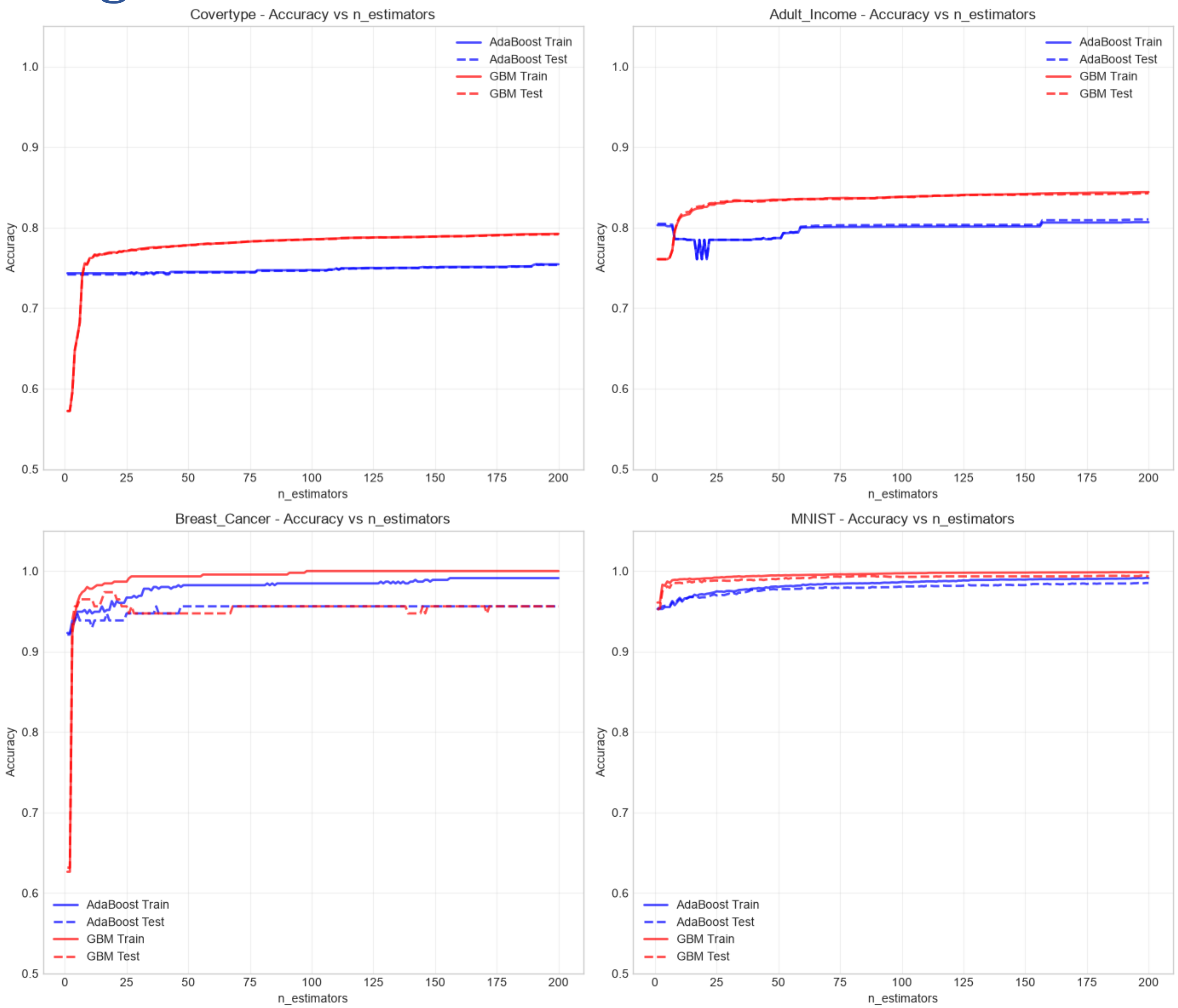




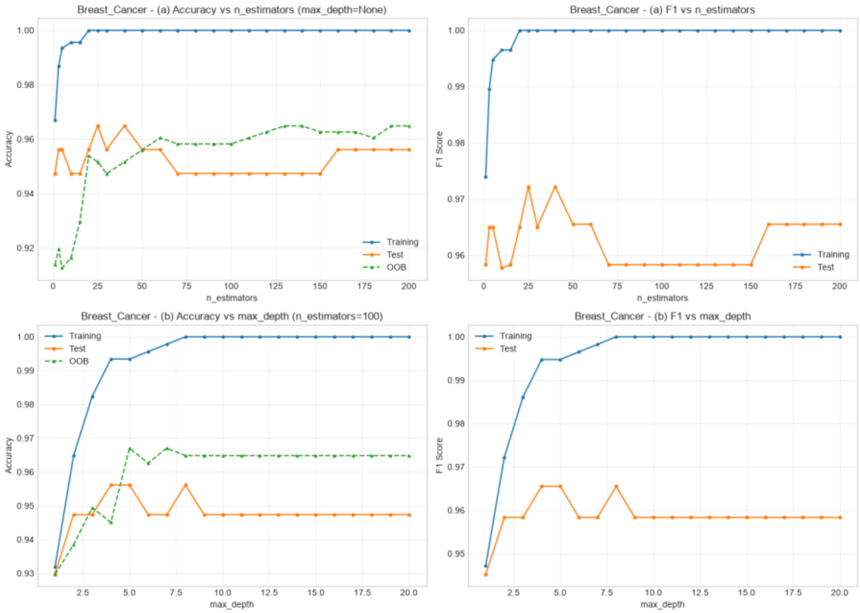


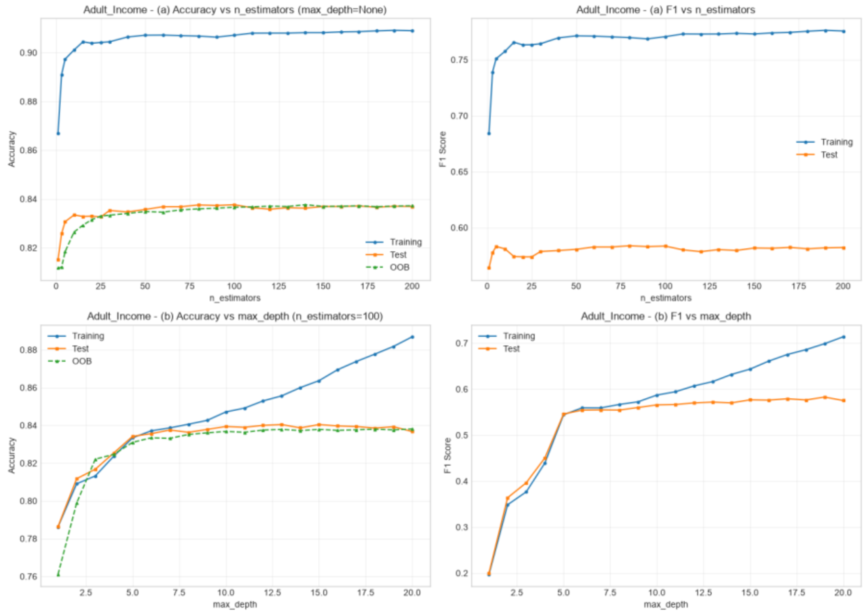


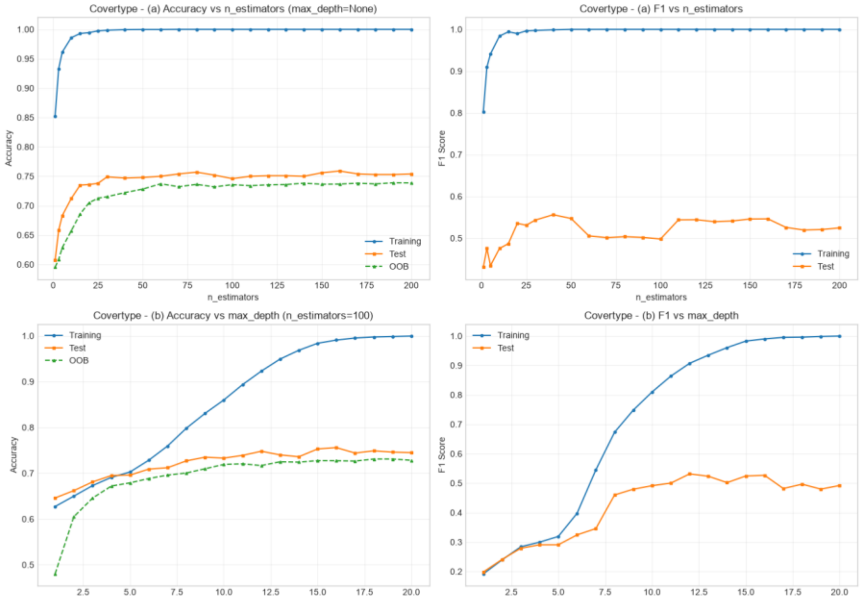
Gradient Boosting Machine

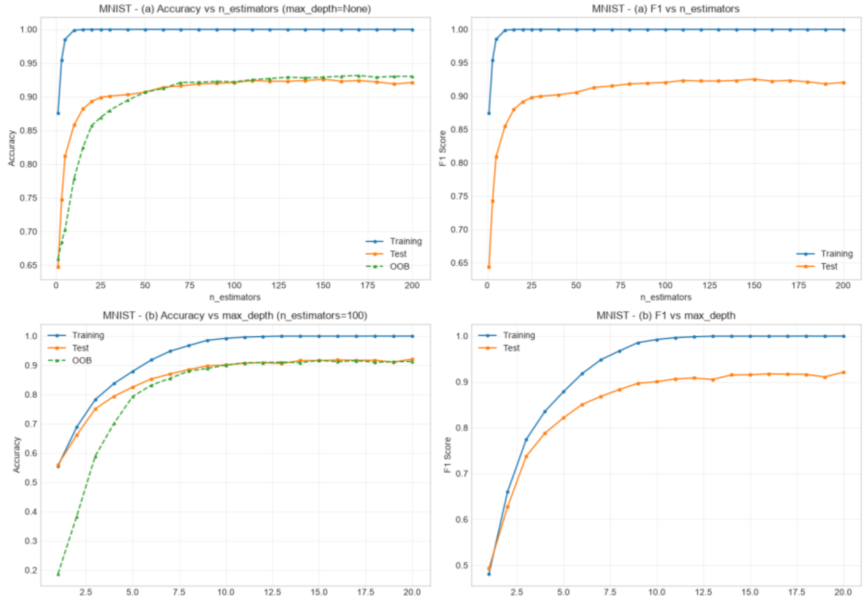


Random Forest Scaling









Head-to-head Comparison

Adult Income

#	Model	Accuracy	Macro F1	AUC–ROC
0	Single Tree	0.7737 ± 0.0040	0.6920 ± 0.0054	0.6942 ± 0.0059
1	AdaBoost	0.8317 ± 0.0056	0.7228 ± 0.0090	0.8535 ± 0.0083
2	Random Forest	0.8379 ± 0.0040	0.7378 ± 0.0078	0.8587 ± 0.0065
3	sklearn RF	0.8083 ± 0.0028	0.7205 ± 0.0044	0.8245 ± 0.0053

Breast Cancer

#	Model	Accuracy	Macro F1	AUC–ROC
0	Single Tree	0.9104 ± 0.0208	0.9041 ± 0.0227	0.9048 ± 0.0257
1	AdaBoost	0.9666 ± 0.0200	0.9639 ± 0.0217	0.9935 ± 0.0070
2	Random Forest	0.9543 ± 0.0114	0.9509 ± 0.0125	0.9913 ± 0.0080
3	sklearn RF	0.9526 ± 0.0228	0.9489 ± 0.0255	0.9878 ± 0.0096

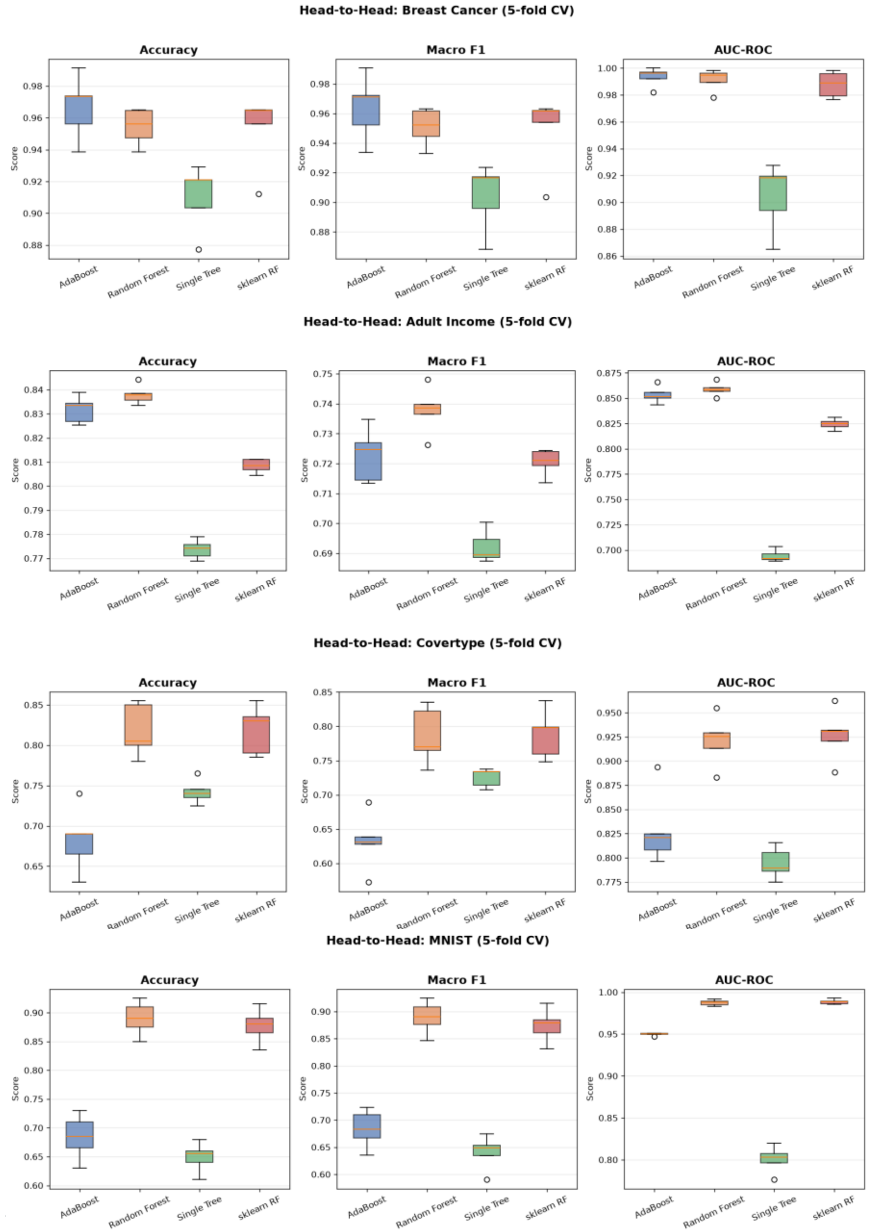
Covertype

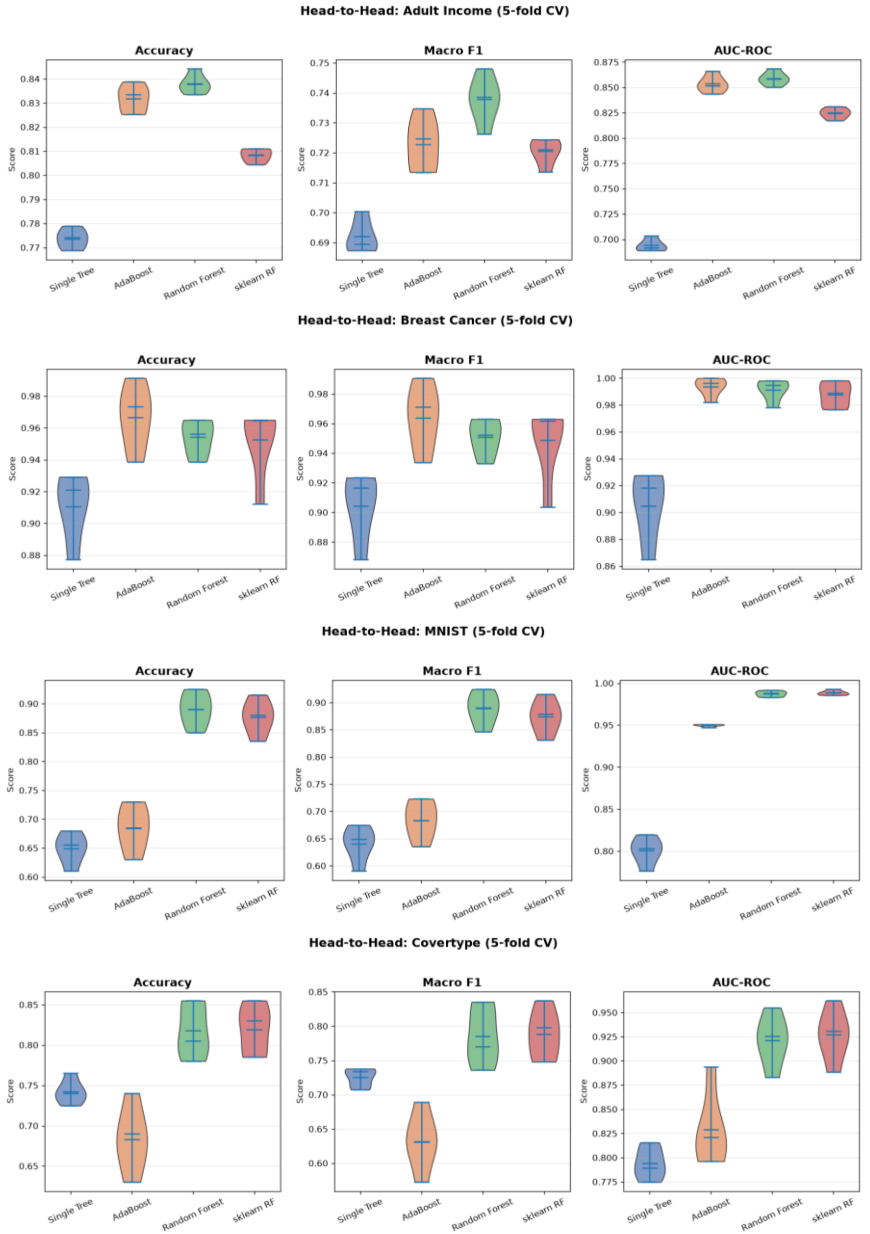
#	Model	Accuracy	Macro F1	AUC–ROC
0	Single Tree	0.7420 ± 0.0148	0.7252 ± 0.0137	0.7942 ± 0.0163
1	AdaBoost	0.6830 ± 0.0402	0.6317 ± 0.0414	0.8287 ± 0.0381
2	Random Forest	0.8180 ± 0.0329	0.7855 ± 0.0418	0.9211 ± 0.0261
3	sklearn RF	0.8190 ± 0.0303	0.7884 ± 0.0357	0.9269 ± 0.0264

MNIST

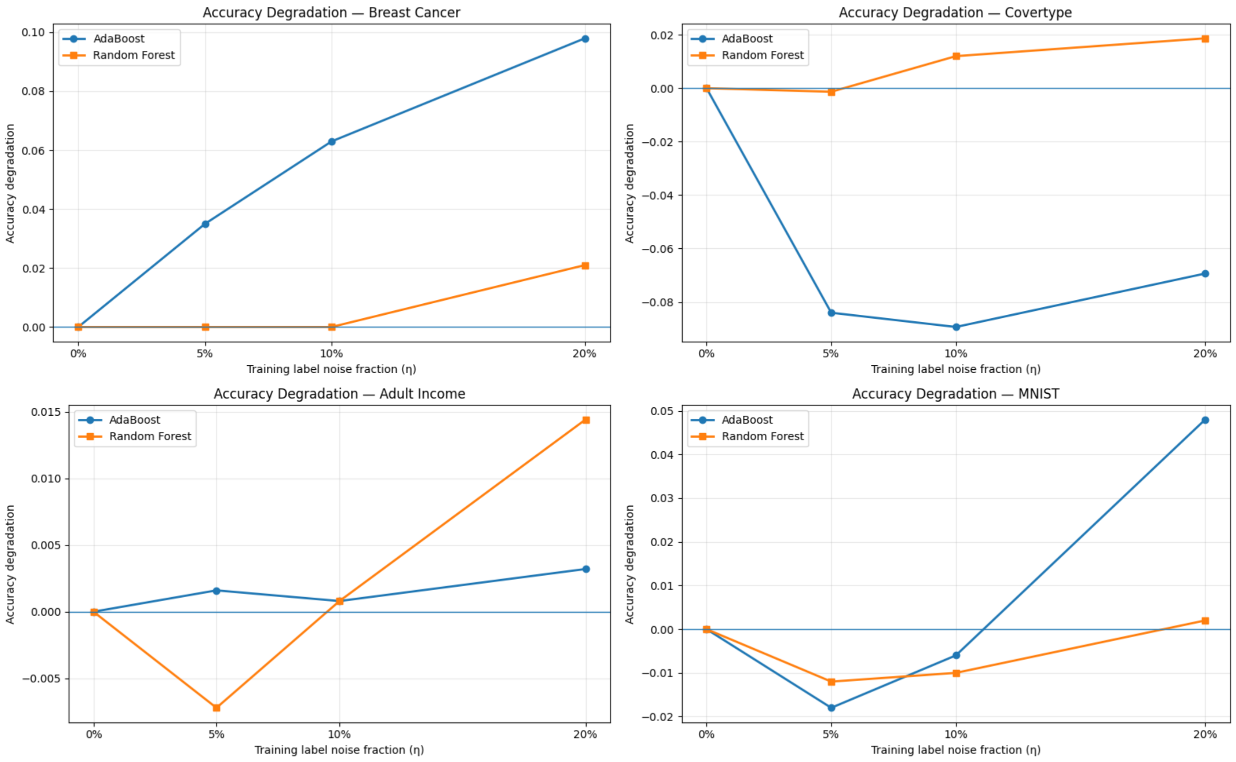
#	Model	Accuracy	Macro F1	AUC–ROC
0	Single Tree	0.6490 ± 0.0261	0.6402 ± 0.0316	0.8005 ± 0.0160
1	AdaBoost	0.6840 ± 0.0390	0.6835 ± 0.0350	0.9498 ± 0.0016
2	Random Forest	0.8900 ± 0.0294	0.8891 ± 0.0301	0.9874 ± 0.0034
3	sklearn RF	0.8770 ± 0.0297	0.8741 ± 0.0312	0.9884 ± 0.0028

Head-to-head Comparison





Noise Robustness



Model Sensitivity to Label Noise

#	Dataset	Mean AdaBoost Degradation	Mean Random Forest Degradation	More Sensitive Model
1	Breast Cancer	0.065268	0.006993	AdaBoost
2	Adult Income	0.001867	0.002667	Random Forest
3	Covertype	-0.080889	0.009778	Random Forest
4	MNIST	0.008000	-0.006667	AdaBoost

Degradation = (Accuracy without noise) − (Average Accuracy with noise at η = 5%, 10%, 20%).
Higher positive values indicate greater performance drop and thus higher sensitivity to label noise.

Unsupervised Analysis

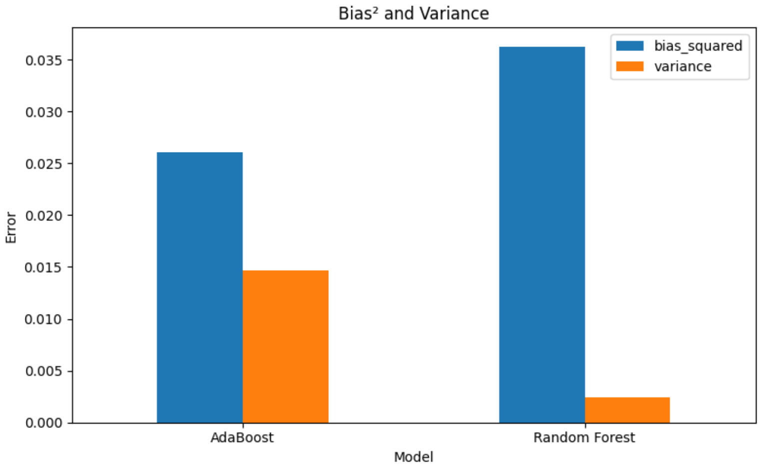
2D PCA scatter plots colored by true class, K-Means clusters, and DBSCAN clusters.

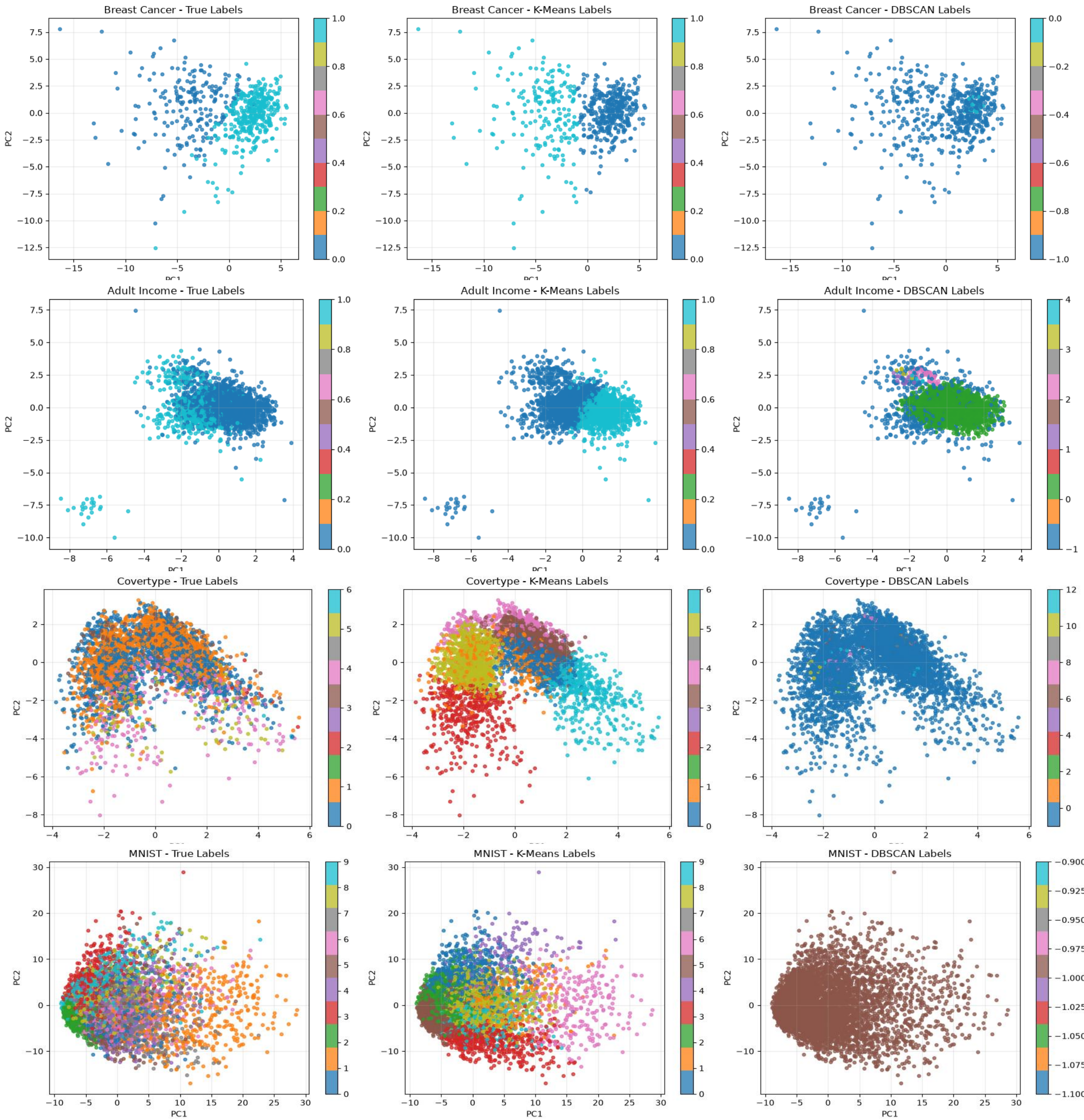
Bias-variance Decomposition

Bias–Variance Analysis — Breast Cancer Dataset

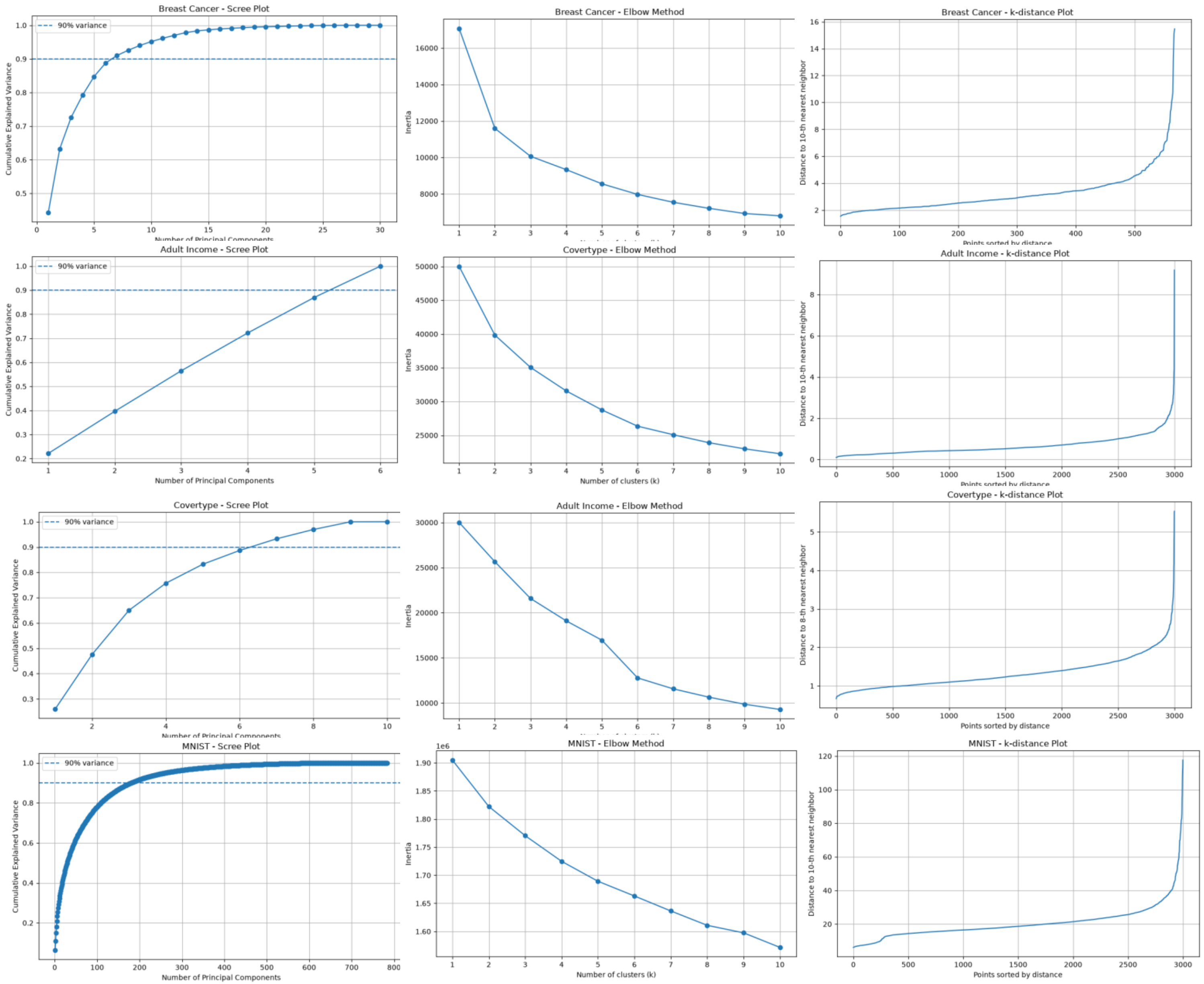
Metric	AdaBoost	Random Forest
Bias²	0.026062	0.036277
Variance	0.014658	0.002404
Total error	0.040720	0.038681
Bias² + Variance	0.040720	0.038681
Decomposition difference	0	0
Mean individual accuracy (Average accuracy of individual models)	0.9556	0.9415
Aggregated accuracy (Accuracy of the aggregated prediction)	0.9649	0.9474

All values are computed on the Breast Cancer dataset using 100 resampled datasets.





Unsupervised Analysis



PCA Variance Explained Summary

Dataset	PC1 Variance	PC2 Variance	First 2 PCs Total Variance	# of PCs for ≥ 90% Variance
Breast Cancer	0.4427	0.1897	0.6324	7
Adult Income	0.2217	0.1750	0.3967	6
Covertype	0.2602	0.2155	0.4757	7
MNIST	0.0632	0.0450	0.1082	180

PC1 Variance and **PC2 Variance** represent the proportion of total variance explained by the first and second principal components.
First 2 PCs Total Variance is the cumulative variance explained by the first two components.
# of PCs for ≥ 90% Variance is the minimum number of principal components needed to explain at least 90% of the total variance.

K-Means Clustering Summary (k = 1–10)

Dataset	Best ARI ↑	Best k (by ARI)
Breast Cancer	0.6536 (≈ 0.6536246)	2
Adult Income	0.0890 (≈ 0.0890122)	4
Covertype	0.0411 (≈ 0.0411467)	5
MNIST	0.2813 (≈ 0.2812668)	10

Best ARI indicates the highest Adjusted Rand Index achieved across k = 1–10, while Best k denotes the corresponding number of clusters.

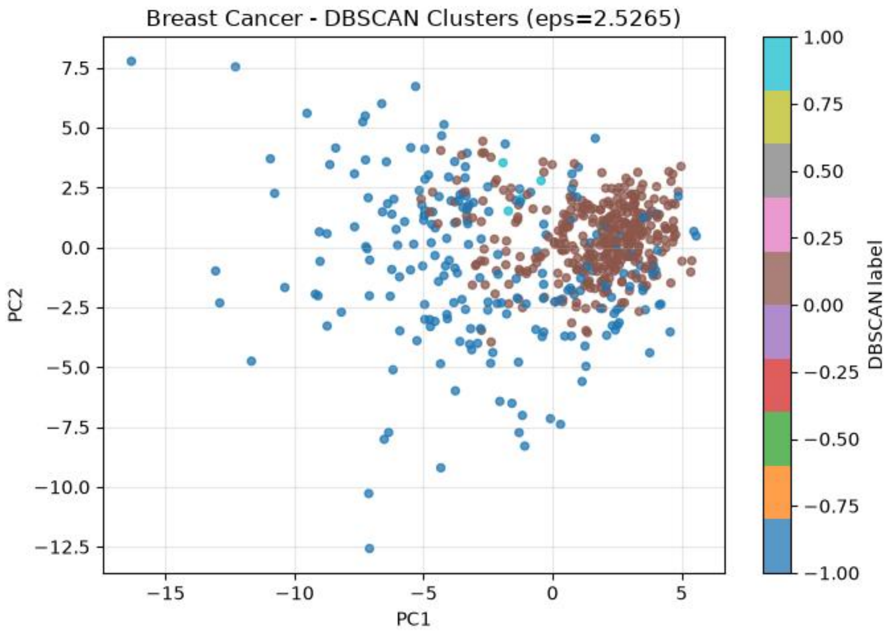
DBSCAN Clustering Summary

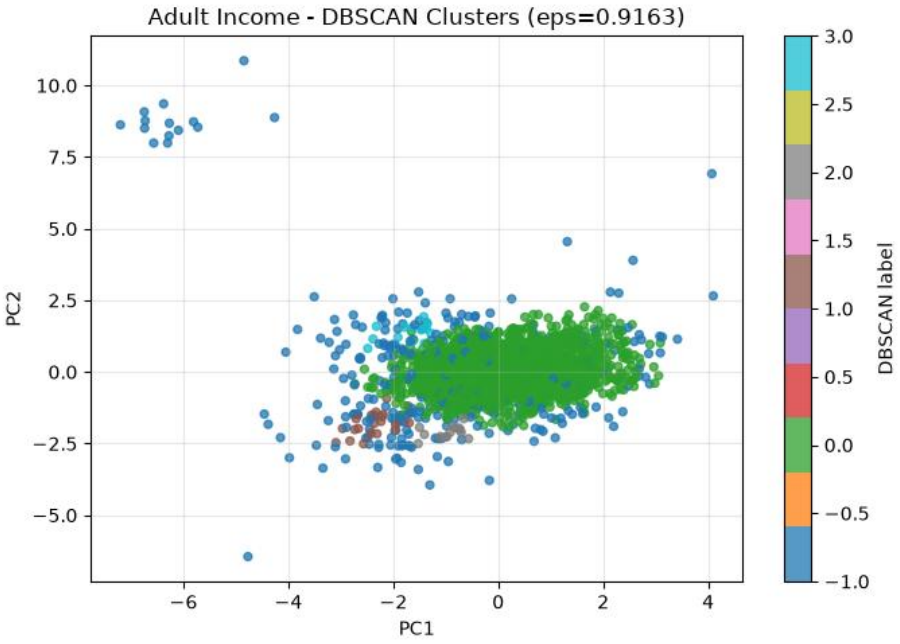
Dataset	Best ε (Epsilon)	Best ARI (Adjusted Rand Index)	Clusters (Excluding Noise)	Noise Fraction (Proportion of Noise Points)
Breast Cancer	2.5265	0.271465 (≈ 0.2714646)	2	0.4587 (≈ 0.458699)
Adult Income	0.9163	0.153203 (≈ 0.1532030)	4	0.1080 (≈ 0.1080)
Covertype	1.2788	0.049041 (≈ 0.0490406)	6	0.2507 (≈ 0.250667)
MNIST	15.0823	0.059434 (≈ 0.0594342)	2	0.6597 (≈ 0.659667)

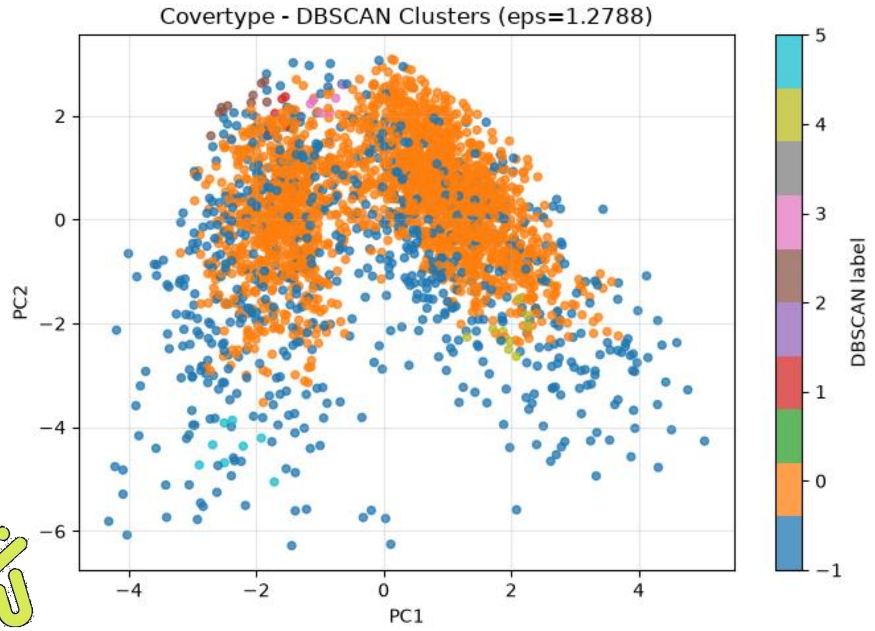
Best ε is the epsilon value that yields the highest Adjusted Rand Index (ARI).
Noise Fraction represents the proportion of points labeled as noise (label = −1) by DBSCAN.

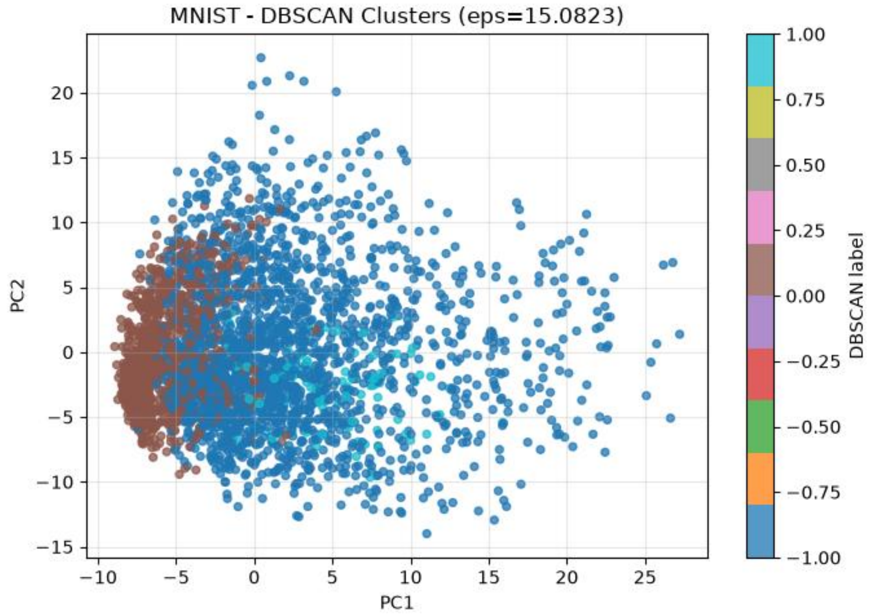
Unsupervised Analysis

DBSCAN Plot









Limitations

- Custom implementations are slower than optimized libraries such as scikit-learn.
- MNIST require high computational time and memory.
- Limited RAM constrained the size of datasets that could be processed in a single run.
- Some experiments used subsets of large datasets to avoid memory overflow and excessive runtime.
- Hyperparameter tuning was limited due to computational cost.
- PCA performs only linear dimensionality reduction and may miss nonlinear structures.

Thank You — Questions?